

Heavy metals in wild rice from northern Wisconsin.



Wild rice grain samples from various parts of the world have been found to have elevated concentrations of heavy metals, raising concern for potential effects on human health. It was hypothesized that wild rice from north-central Wisconsin could potentially have elevated concentrations of some heavy metals because of possible exposure to these elements from the atmosphere or from water and sediments. In addition, no studies of heavy metals in wild rice from Wisconsin had been performed, and a baseline study was needed for future comparisons. Wild rice plants were collected from four areas in Bayfield, Forest, Langlade, Oneida, Sawyer and Wood Counties in September, 1997 and 1998 and divided into four plant parts for elemental analyses: roots, stems, leaves and seeds. A total of 194 samples from 51 plants were analyzed across the localities, with an average of 49 samples per part depending on the element. Samples were cleaned of soil, wet digested, and analyzed by ICP for Ag, As, Cd, Cr, Cu, Hg, Mg, Pb, Se and Zn. Roots contained the highest concentrations of Ag, As, Cd, Cr, Hg, Pb, and Se. Copper was highest in both roots and seeds, while Zn was highest just in seeds. Magnesium was highest in leaves. Seed baseline ranges for the 10 elements were established using the 95% confidence intervals of the medians. Wild rice plants from northern Wisconsin had normal levels of the nutritional elements Cu, Mg and Zn in the seeds. Silver, Cd, Hg, Cr, and Se were very low in concentration or within normal limits for food plants. Arsenic and Pb, however, were elevated and could pose a problem for human health. The pathway for As, Hg and Pb to the plants could be atmospheric.